# Environmental Engineering

## Unit V: Sewage Treatment and Disposal

**Syllabus**

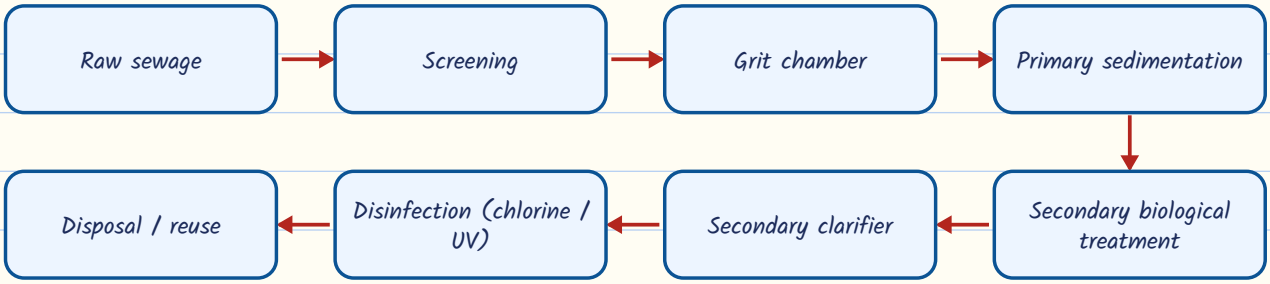
Objectives - Selection of treatment methods - Principles, functions - Activated sludge process and extended aeration systems - Trickling filters - Sequencing batch reactor (SBR) - UASB - Waste stabilization ponds - Other treatment methods - Reclamation and reuse of sewage - Recent advances in sewage treatment - Construction, operation and maintenance aspects - Discharge standards - Sludge treatment - Disposal of sludge.

## 1. Objectives of Sewage Treatment

- To remove **suspended solids, organic matter (BOD)** and pathogens before disposal
- To **protect receiving waters** (rivers, lakes, sea) from oxygen depletion and pollution
- To prevent **public health hazards** and nuisances (smell, flies)
- To produce effluent suitable for **reuse** (irrigation, industry, ground water recharge)
- To recover **resources**: biogas, manure, treated water, nutrients

### Levels of treatment

| Level | Units | Removes |
|---|---|---|
| **Preliminary** | Screens, grit chamber, skimming tank, comminutor | Floating matter, grit, oil and grease |
| **Primary** | Primary sedimentation tank | Settleable solids; **60% SS, 30-35% BOD** |
| **Secondary (biological)** | Activated sludge, trickling filter, ponds, SBR, UASB | Dissolved and colloidal organics; **85-95% BOD** |
| **Tertiary / advanced** | Nutrient removal, filtration, chlorination, UV, RO | Nitrogen, phosphorus, pathogens, residual organics |



**Selection of treatment methods depends on**

* Characteristics of the sewage and **required effluent standards**
* **Land availability** (ponds need big area; ASP is compact) and cost of land
* **Capital cost, operating cost and power** needs (trickling filters and ponds use less power)
* Skilled operators, climate (ponds and UASB work best in warm climate), reuse of by-products (biogas)
* Future expansion, variations in flow and load (shock loads)

## 2. Preliminary Treatment

- **Screens** - coarse (bars 25-50 mm apart), medium (12-25 mm), fine (6-12 mm); manual or mechanical; velocity through bars 0.6-1 m/s; screenings are buried or incinerated
- **Comminutors** - cut the solids to small pieces, so that they can be carried in the flow
- **Grit chamber** - removes inorganic grit (sand, gravel, 0.2 mm, S.G. 2.65) to protect pumps from wear; **horizontal flow velocity = 0.3 m/s**, detention time 30-60 s; types: constant velocity (parabolic channel, proportional weir), **aerated grit chamber**, vortex
- **Skimming tanks** - remove oil and grease by floating them on the surface with air bubbles; detention 3 min
- **Equalisation tank** - balances flow and load variations

## 3. Primary Sedimentation

- Settles the organic suspended solids to get **raw primary sludge** (3-5% solids)
- Rectangular (L:B = 3-5:1) or circular (diameter 20-40 m, side-water depth 3-4 m); **detention time 1.5 – 2.5 h**; surface loading **25 – 40 $m^3/m^2/day$** (up to 50 at peak); weir loading 125-250 $m^3/m/day$
- Mechanical scrapers move sludge to hoppers; **Imhoff tank** (two-storey: sedimentation + digestion) is for small plants
- Removes 60-70% SS and 30-35% BOD

# 4. Biological Treatment Principles

Micro-organisms (bacteria, protozoa) consume the dissolved organic matter and use it for **energy and growth**; the cell mass (sludge) is then separated. **Aerobic** processes use oxygen, **anaerobic** work without oxygen (producing $CH_4$).

### Aerobic oxidation

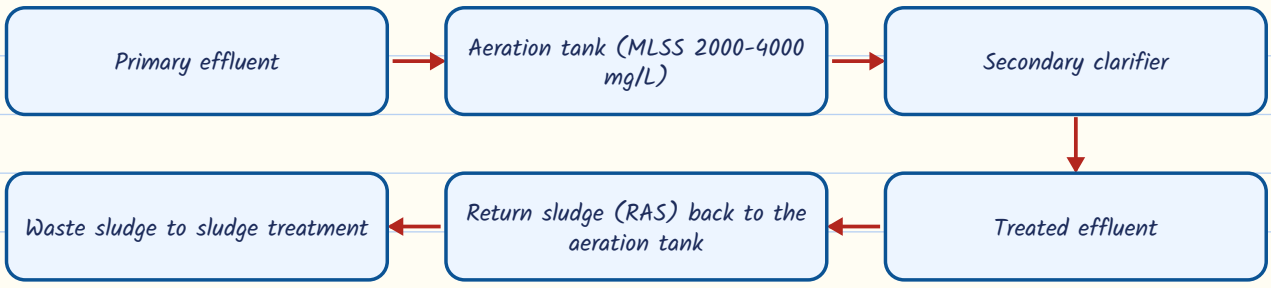
Organic matter + $O_2$ + bacteria -> $CO_2$ + $H_2O$ + new cells + energy

Anaerobic: organic matter -> organic acids -> **$CH_4$ + $CO_2$** (methanogens)

| Growth type | Description |
|---|---|
| **Suspended growth** | Micro-organisms in suspension in the aeration tank (activated sludge, ponds, SBR) |
| **Attached growth** | Biofilm on a medium: trickling filters, RBC, MBBR |
| **Aerobic** | Needs DO of 1.5-2 mg/L |
| **Anaerobic** | UASB, digesters, septic tank, Imhoff tank |

# 5. Activated Sludge Process (ASP)

The most widely used **suspended growth aerobic process**. Settled sewage is mixed with a mass of micro-organisms (**activated sludge**) in an **aeration tank**, and air supplied. The mixed liquor flows to a **secondary clarifier** where sludge settles; a part of the **settled sludge is recycled** (return activated sludge, RAS) and the rest wasted (WAS).



## Important parameters

| Parameter | Definition / typical value |
|---|---|
| **MLSS** | Mixed liquor suspended solids; 2000 - 4000 mg/L (conventional) |
| **MLVSS** | Volatile fraction (about 70-80% of MLSS) |
| **F/M ratio** | Food to micro-organism ratio = Q $BOD_0$ / (V X) = kg BOD per day / kg MLSS; **0.2 - 0.5 per day** (conventional); extended aeration 0.05-0.15 |
| **HRT (theta)** | V / Q = 4 - 8 h (conventional) |
| **SRT / MCRT ($theta_c$)** | Sludge age = V X / ($Q_w X_r + Q_e X_e$) = 5 - 15 days |
| **SVI** | Sludge volume index = (30-min settled volume in mL/L) / (MLSS in g/L); good settling **50 - 150 mL/g**; above 150 indicates bulking |
| **Recycle ratio R** | $Q_r/Q$ = X / ($X_r$ - X) = 0.25 - 1.0 |
| **Volumetric loading** | 0.3 - 0.7 kg BOD/$m^3$/day |
| **Oxygen requirement** | About 1 - 1.2 kg $O_2$ per kg BOD removed |
| **BOD removal efficiency** | 85 - 95% |

### Design equations

$V = theta_c\ Y\ Q\ (S_0 - S) / [X\ (1 + k_d\ theta_c)]$ (Y = yield 0.4-0.8, $k_d$ = decay 0.025-0.075 per day)

Sludge produced: $P_x = Y\ Q\ (S_0 - S) / (1 + k_d\ theta_c)$

Efficiency $E = (S_0 - S) / S_0 \times 100$

### Example

Q = 5 MLD, $BOD_0$ = 150 mg/L (after primary), MLSS = 3000 mg/L, F/M = 0.3. BOD load = 5000 x 150 /1000 = 750 kg/d.

V = 750/(0.3 x 3) = 833 $m^3$ => HRT = 833/5000 x 24 = 4 h. With depth 4 m, area = 208 $m^2$.

## Aeration systems

- **Diffused aeration** - fine bubble diffusers (membrane, ceramic) at bottom, air from blowers; high oxygen transfer efficiency
- **Mechanical (surface) aerators** - low-speed vertical turbines or brush aerators
- Mixing needs and oxygen demand both govern; **DO 1.5 - 2 mg/L** maintained

## Modifications of ASP

| Modification | Key feature |
|---|---|
| **Conventional (plug flow)** | Uniform sewage input at one end; F/M 0.2 – 0.4 |
| **Step aeration** | Sewage fed in steps along the tank; uniform oxygen demand |
| **Tapered aeration** | Air supply decreases along length |
| **Complete mix** | Inlet and outlet distributed; resistant to shock loads |
| **Contact stabilisation** | Short contact and separate re-aeration of RAS |
| **Extended aeration** | Long sludge age, low F/M, see below |
| **Oxidation ditch** | Racetrack channel with brush rotors |
| **High-rate ASP** | High F/M 1.5 – 5, short HRT 0.5 – 2 h |

## Operational problems

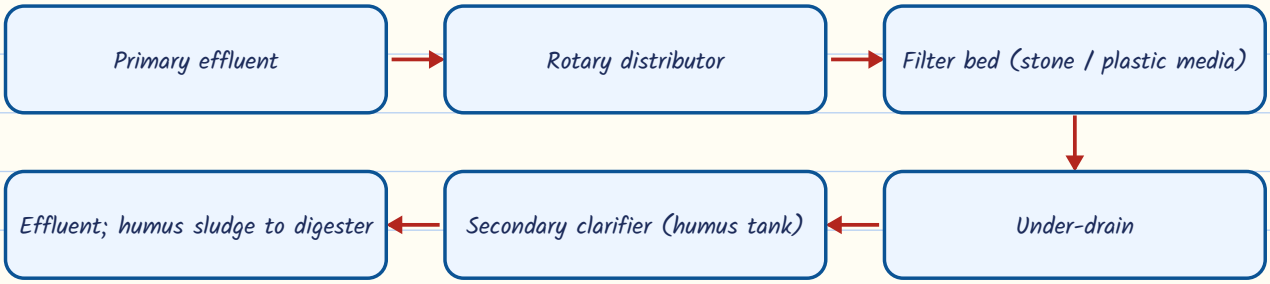
- **Sludge bulking** – filamentous bacteria (Sphaerotilus, Nocardia), SVI above 150, poor settling; remedy: chlorinate return sludge, adjust F/M, increase DO
- **Rising sludge** (denitrification in clarifier) and **foaming**
- **Pin floc**, **toxic shock** loads, **low DO**

# 6. Extended Aeration Systems

- A variant of ASP where sewage is aerated for a **long time (18 – 36 h)**, with **low F/M (0.05 – 0.15)** and a high sludge age **(20 – 30 days)**
- Micro-organisms start **endogenous respiration** (they consume each other, so sludge is **stabilised aerobically**); the excess sludge is small and fairly stable, needs no digestion
- **No primary settling** needed (raw screened sewage after grit); MLSS 3000-6000 mg/L; aeration tank volume large; **oxidation ditch** is an example
- Suited to **small communities, hotels, hostels, packaged plants**; simple operation; **BOD removal 90-95%**, nitrification occurs
- Disadvantages: **high power cost (aeration)**, big tank, high MLSS in the effluent solids

# 7. Trickling Filters

A bed of **coarse media (stones, plastic, slag)** over which settled sewage is sprayed or distributed intermittently. A **biofilm (zoogleal film)** grows on the surface of the media; as sewage trickles down, the organics are absorbed and oxidised, with **air flowing naturally** through voids.




- **Components:** circular tank, **rotary distributor** (hydraulic arms), **filter medium** (stone 40-80 mm, depth 1.8-2.5 m; plastic modules 4-12 m), **under-drainage** (with ventilation), **humus tank** (secondary settling)
- Sloughing of the biofilm (humus) is removed in the humus tank

| Parameter | Low (standard) rate | High rate |
|---|---|---|
| Hydraulic loading | 1 - 4 $m^3/m^2/day$ | 10 - 40 $m^3/m^2/day$ |
| Organic loading | 80 - 320 g $BOD/m^3/day$ (0.08-0.32 kg) | 320 - 1000 g $BOD/m^3/day$ |
| Depth | 1.8 - 3 m | 1 - 2 m (stone), up to 12 m plastic |
| Recirculation | None | **1 - 4 times** |
| BOD removal | 80 - 90% | 65 - 80% |
| Effluent | Nitrified, stable | Not nitrified |
| Problems | Filter flies (Psychoda), odour | Ponding, odour |

### NRC formula (high rate, single stage)

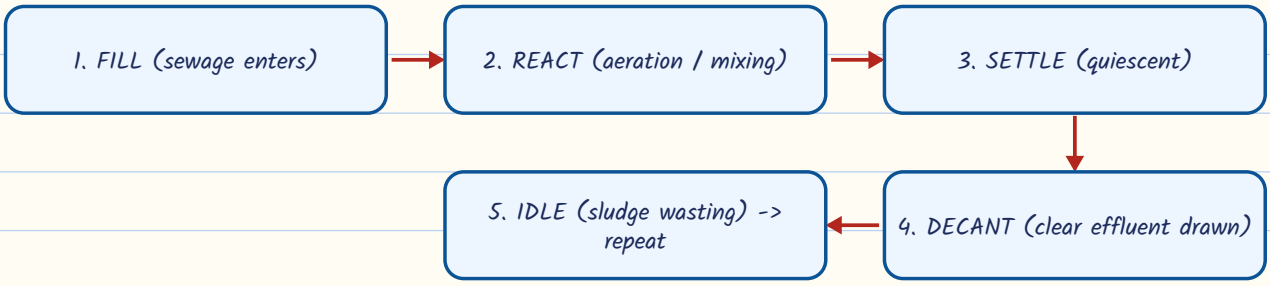
$E = 100 / (1 + 0.4432\ sqrt(W / (V F)))$ ($W$ = BOD load in kg/day, $V$ = volume in $m^3$, $F$ = recirculation factor = $(1 + R)/(1 + 0.1R)^2$)

- **Merits:** low power, simple, resists shock loads, low sludge; **demerits:** big land, odour and flies, clogging, lower efficiency in cold

- **Related attached-growth processes: Rotating biological contactor (RBC)** - discs rotating partly in sewage; **Biotowers**; **Moving bed bio-reactors (MBBR)**; **Submerged aerated filters**

## 8. Sequencing Batch Reactor (SBR)

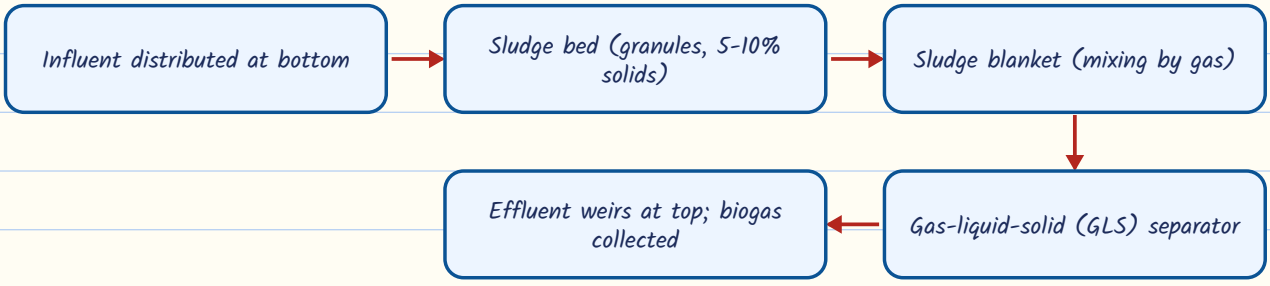
The SBR is a **fill-and-draw activated sludge process** in which all steps - aeration and clarification - take place in **a single tank** in a time sequence. No separate clarifier and no sludge return pump are needed.




- Cycle 4 - 8 h (fill 1-2 h, react 2-4 h, settle 0.5-1 h, decant 0.5-1 h); **at least two tanks** work alternately for continuous inflow
- **Decanter** (floating or fixed) removes treated water; **PLC / SCADA** controls the cycle
- **Merits:** compact, flexible, good **nutrient removal (nitrogen and phosphorus)** by varying aerobic and anoxic phases, no sludge bulking issues, no secondary clarifier
- **Demerits:** needs **automatic control and skilled maintenance**, peak flow storage, decanter issues. Variants: **CASS, ICEAS, MBBR-SBR**

## 9. UASB (Upflow Anaerobic Sludge Blanket) Reactor

An **anaerobic** treatment unit in which sewage flows **upward** through a **dense blanket of granular sludge** at the reactor bottom. Anaerobic bacteria break the organics into **biogas (60-70% methane)**. Developed in the Netherlands (Lettinga); widely used in India (Karnal, Kanpur, Mirzapur under the Ganga Action Plan).



- **Design parameters:** HRT **6 – 8 h** (domestic sewage), upflow velocity **0.5 – 1.0 m/h**, height 4 – 6 m, **OLR 1 – 3 kg COD/$m^3$/day**, temperature 20-35 degrees C
- **BOD removal 60 – 80%**; produces **biogas (energy)**, and **little sludge (well stabilised)**; low power; low cost; the effluent needs **post-treatment** (polishing pond, ASP, or aerated lagoon) to meet standards
- **Demerits:** slow start-up (months, seeding needed), odour ($H_2S$), sensitive to toxic shocks and low temperature, effluent with high COD and pathogens

# 10. Waste Stabilization Ponds (WSP)

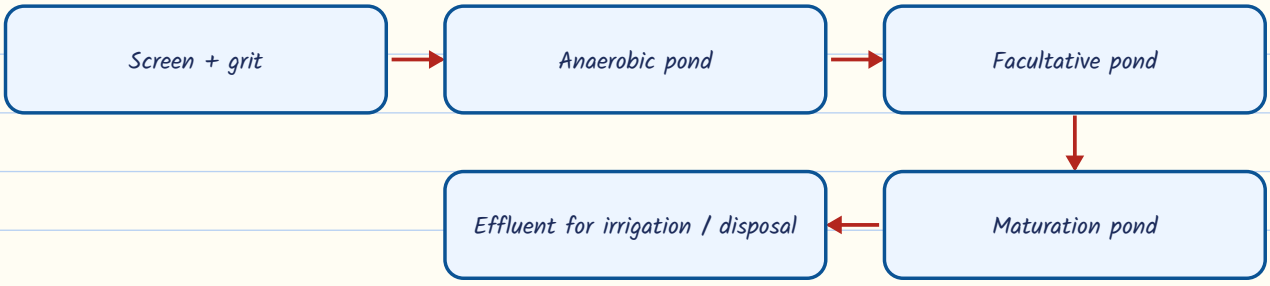
Shallow earthen basins in which sewage is treated by **natural processes** (sunlight, algae, bacteria, wind). Algae produce oxygen by **photosynthesis** for bacteria that oxidise organic matter; this **symbiosis** is the principle.

### Algal-bacterial symbiosis

Organic matter + $O_2$ (from algae) --bacteria--> $CO_2$ + $NH_3$ + $PO_4$ + new cells

$CO_2$ + nutrients + sunlight --algae--> new algae + $O_2$

| Pond | Depth | Detention | Features |
|---|---|---|---|
| **Anaerobic** | 2.5 – 5 m | 2 – 5 d | Heavy BOD (OLR up to 400 g/$m^3$/d), sludge digests, 50-60% BOD removal, odour |
| **Facultative** | 1.0 – 1.5 m | 5 – 30 d | Aerobic top, anaerobic bottom; most common; BOD loading 100-400 kg/ha/d |
| **Maturation (aerobic)** | 0.8 – 1.2 m | 5 – 10 d | Removes **pathogens (bacteria)**, N, with UV; low BOD feed |
| **Aerated lagoon** | 2 – 4 m | 3 – 10 d | Mechanical aerators, higher loads |




- **Merits:** cheap, **no power**, simple O and M, high pathogen removal; **demerits: large land**, odour in anaerobic ponds, mosquitoes, algal in effluent (SS), seepage
- **Design:** area from BOD loading (kg/ha/day) = BOD x Q / area; e.g., 1 MLD at 200 mg/L is 200 kg/d; at 200 kg/ha/d needs 1 ha

- **Fish and aquatic (duckweed / water hyacinth) ponds** are also used for polishing

# 11. Other Treatment Methods

| Method | Notes |
|---|---|
| **Septic tank + soak pit** | On-site, for houses; detention 24 h; digestion of settled sludge, 30-50% BOD removal; effluent to soak pit or dispersion trench |
| **Imhoff tank** | Settling above and digestion below |
| **Oxidation ditch** | Extended aeration, racetrack; brush rotors |
| **Rotating biological contactor (RBC)** | Discs 40% submerged, 1-2 rpm |
| **Anaerobic filter, Anaerobic baffled reactor** | Small plants |
| **Constructed wetlands / root zone** | Reeds and gravel; decentralised, natural |
| **Membrane bio-reactor (MBR)** | ASP + membrane filtration, no clarifier; excellent effluent |
| **Moving bed biofilm reactor (MBBR)** | Plastic carriers in suspended aeration tank; compact |
| **Fluidised bed reactor** | Biofilm on sand carriers |

# 12. Reclamation and Reuse of Sewage

With shortage of fresh water, treated sewage is a source of water. **Reclamation** = treating the wastewater to a quality fit for a beneficial use; **reuse** is the use.

| Reuse | Required treatment |
|---|---|
| **Agriculture / irrigation** (non-food crops, orchards, with food crops need higher quality) | Secondary + disinfection; sewage farming |
| **Industrial** (cooling water, boilers, process) | Secondary + filtration + softening |
| **Urban non-potable** (toilet flushing, gardens, fire fighting, road washing, car wash) | Secondary + filtration + disinfection |
| **Ground water recharge** (injection, spreading basins) | Tertiary + RO |
| **Recreational lakes, aquaculture (fish ponds)** | Secondary + maturation / disinfection |
| **Indirect / direct potable reuse** | Advanced: MF - RO - UV - AOP (e.g., NEWater Singapore, Windhoek) |

- **Risks:** pathogens, heavy metals, salts, emerging contaminants (pharmaceuticals); need strict **monitoring and standards**, public acceptance

- **Benefits:** water saving, nutrients (N, P) for crops, less pollution load, drought-proof supply, reduced fresh water demand
- Policy: **National Water Policy** and Urban Wastewater reuse policies encourage use of treated sewage by industries and for irrigation

## 13. Recent Advances in Sewage Treatment

- **Membrane bioreactors (MBR)** and **Anaerobic MBR**
- **Aerobic granular sludge (Nereda)** - compact, nutrient removal, no clarifier
- **Anammox (anaerobic ammonium oxidation)** - cheaper nitrogen removal; **enhanced biological phosphorus removal (EBPR); A2O / Bardenpho / modified UCT processes**
- **Advanced oxidation processes (AOP):** ozone, UV/$H_2O_2$ for micropollutants
- **Constructed wetlands, phytoremediation, decentralised systems (DEWATS)**
- **Microbial fuel cells, energy-positive plants** (biogas CHP), **resource recovery** (struvite, phosphorus)
- **Smart operation:** online sensors, **SCADA / AI-based control**; **Hybrid Annuity Model (HAM)** plants under **Namami Gange**
- **Nano-filtration, forward osmosis, electrocoagulation**, and **solar-powered treatment**

## 14. Construction, Operation and Maintenance Aspects

- **Construction:** units on **hydraulic gradient**, preferably gravity flow, in water-tight RCC; **bypass** and flow measurement (Parshall flume), standby units; plant located **downstream and away** from the town (buffer zone), with flood protection and room for expansion
- **Operation:** monitor flow, DO, MLSS, SVI, F/M, pH, BOD, COD, SS, residual chlorine, and keep **log sheets**; control sludge wasting and recycle; adjust air supply
- **Maintenance:** clean screens daily, remove grit, lubricate gearboxes and scrapers, clean diffusers, repaint, desludge ponds, calibration; **preventive maintenance schedule**, spare parts
- **Safety and health:** toxic gases ($H_2S$, $CH_4$), confined spaces, pathogens (vaccination, PPE), **odour control** (bio-filters, covering); **trained staff and laboratory**

# 15. Discharge Standards (Effluent Standards)

Treated sewage must meet standards of the **Central Pollution Control Board (CPCB)** under the **Environment (Protection) Act, 1986** (and **Water Act, 1974**), before discharging to inland surface water, land for irrigation or marine areas.

| Parameter | Inland surface water | Land for irrigation |
|---|---|---|
| pH | 5.5 - 9.0 | 5.5 - 9.0 |
| $BOD_5$ at 20 degrees C | 30 mg/L (STPs: 10-20 mg/L in new NGT standards) | 100 mg/L |
| COD | 250 mg/L (new STP norms: 50-100) | — |
| Total suspended solids | 100 mg/L (STP: 20-50) | 200 mg/L |
| Oil and grease | 10 mg/L | 10 mg/L |
| Ammonical nitrogen | 50 mg/L (STP: 5) | — |
| Total nitrogen | —(STP: 10) | — |
| Fecal coliform | Less than 1000 MPN/100 mL (desirable 230) | — |
| Temperature | Shall not exceed 5 degrees C above receiving water | — |

Note: The **National Green Tribunal (2019)** and the **CPCB (2017)** have set stricter norms for STPs: BOD below 10 mg/L, TSS 20, FC 100 MPN/100 mL in sensitive areas (values vary by capacity and location; check the latest notification).

# 16. Sludge Treatment

Sludge is a mixture of solids and water (**98–99% water** in primary sludge, 99–99.5% in activated sludge). Its treatment aims to reduce volume, stabilise organics, destroy pathogens and make it fit for disposal.



## Sludge volume and moisture

### Relation

$V_1/V_2 = (100 - P_2) / (100 - P_1)$ (P = % moisture)

Example: 100 $m^3$ sludge at 98% moisture reduces to 100 x 2/5 = **40 $m^3$** at 95%.

**Sludge Volume Index (SVI)** - settling characteristic of activated sludge

## Anaerobic digestion

- Stabilises sludge in closed tanks without oxygen; **temperature 35 degrees C (mesophilic)**, retention 20-30 days (high rate 10-15 d); pH 6.8-7.2
- **Stages:** (1) **hydrolysis**, (2) **acid formation (acidogenesis)**, (3) **methanogenesis (methane formation)**
- **Biogas** - 60-65% $CH_4$, 30-35% $CO_2$, traces of $H_2S$; **0.8 - 1.1 $m^3$ gas per kg volatile solids destroyed**; used for heating the digester or for power generation
- **Types:** standard rate (unheated, no mixing), **high-rate (heated, mixed)**, **two-stage** (primary + secondary)
- **Digester gas** is collected in gasholder; **digested sludge** is dark, inoffensive and with 50-60% volatile reduction
- **Operational problems:** foaming, scum formation, pH drop (sour digester), toxicity

## Conditioning and dewatering

- **Conditioning:** chemical (alum, ferric chloride, lime, **polyelectrolytes**), heat treatment, elutriation
- **Sludge drying beds** - open sand beds (15-30 cm sand + 20-30 cm gravel), 7-15 days drying, solid content 30-40%, cheapest; needs area (0.1 - 0.25 $m^2$/person); covered in monsoon
- **Mechanical: centrifuge, belt filter press, filter press, vacuum filter** - 15-35% solids, used in big plants
- **Thermal drying and incineration**, **composting**, **vermi-stabilisation**

# 17. Disposal of Sludge

| Disposal method | Notes |
|---|---|
| **Land application / manure** | Digested, dried sludge as soil conditioner (N, P, organic matter); free from pathogens and heavy metals as per norms |
| **Landfill** (sanitary) | Dried sludge, with solid waste; control leachate and gas |
| **Lagooning** | Sludge in earthen lagoons, natural stabilisation |
| **Incineration** | Reduces volume to ash by burning at 800-1000 degrees C; costly, air pollution control |
| **Composting** | With solid waste, high-quality compost |
| **Use as fuel / brick making / cement kiln** (co-processing) | Resource recovery |
| **Disposal into sea** | Banned or restricted |

## Remember for exam

* Primary removes 30-35% BOD; secondary 85-95%
* ASP: F/M 0.2-0.5, MLSS 2000-4000, SVI 50-150, sludge age 5-15 d; extended aeration F/M 0.05-0.15, 18-36 h
* Trickling filter: biofilm, rotary distributor, high rate with recirculation
* SBR: Fill - React - Settle - Decant - Idle, one tank
* UASB: anaerobic, 6-8 h, biogas 60-70% $CH_4$, needs post-treatment
* WSP: algal-bacterial symbiosis; anaerobic - facultative - maturation
* CPCB standard BOD 30 mg/L (inland water); digestion at 35 degrees C produces biogas

## Important questions

1. Explain the objectives and the selection of sewage treatment methods. Draw the flow diagram.
2. Describe the activated sludge process. Explain F/M ratio, MLSS, SVI and sludge age; discuss sludge bulking.
3. Explain the extended aeration system and compare it with conventional ASP.
4. Describe the working of a trickling filter. Differentiate low-rate and high-rate filters.
5. Explain the sequencing batch reactor and the UASB reactor with neat sketches.
6. Describe waste stabilisation ponds and the algal-bacterial symbiosis.
7. Discuss reclamation and reuse of sewage and the recent advances in treatment.
8. Explain sludge digestion, conditioning, dewatering and disposal of sludge.

9. Write notes on discharge standards and O and M aspects of a sewage treatment plant.